Predicting Coronary Heart Disease from BRFSS Survey Data with Linear and Logistic Models

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***Abstract*—[Abstract: task, data, methods, nested-CV F1 of the final model, main finding (codebook preprocessing matters most; class weighting and regularization do not help an underfitting linear model).]**

I. Introduction

[Task: predict self-reported coronary heart disease (MICHD) from the 2015 BRFSS telephone survey; binary classification evaluated with F1; only NumPy allowed. Contributions and paper outline.]

II. Data and preprocessing

[Person 1, data description: training and test sizes, class balance, missing values, feature types. Numbers from the shared EDA: 328,135 training and 109,379 test respondents, 321 features, 8.8% positive, 44.8% of all values missing, 147 features missing for more than half of the respondents, mostly from optional state modules. Optional: EDA figure.]

The features are coded survey answers. Steps 1 to 3 convert them into meaningful values, following the BRFSS 2015 codebook [1], which we checked for every feature; they do not use the labels.

1) *Special codes.* The codes for "don't know" and "refused", such as 7 and 9, become missing values, and the codes for "none", 88, 888 or 555, become 0. For example, 62% of the respondents report 88 = "no days" of poor physical health, otherwise read as 88 days.
2) *Column removal.* We remove 64 features: 32 describe the survey, such as identifiers and weights, 29 are raw answers that also exist in a cleaned version, such as WEIGHT2 replaced by WTKG3, and 3 are nominal. Two of these give the type of physical activity, with 77 categories, already summarized by the calculated energy cost, intensity and weekly minutes of activity; the third is 99.9% missing.
3) *Nominal features.* Seven features use numbers as category labels, such as state and marital status, and are one-hot encoded.
4) *Imputation and scaling.* **[Person 1: median imputation, standardization, bias column; all statistics learned on training rows only.]**

[Person 1: own improvement, if any.]

Features of the final model. Each feature with missing values gets an indicator equal to 1 when the value is missing. Missingness is informative, because many questions are only asked after a given answer: the age at diabetes diagnosis is missing for respondents without diabetes. All 86 features with 3 to 10 observed values are also one-hot encoded, so that each level of, for instance, general health has its own effect.

III. Methods

A. Evaluation protocol

We use nested cross-validation, so that the data used to choose a model is never used to evaluate it. All methods share five stratified folds, with seed 0.

1) *Outer loop.* Each fold is the test fold once; the other four form the training set.
2) *Inner loop.* Three folds of the training set select the hyperparameters and the threshold $t \in \{0.05, 0.06, \ldots, 0.95\}$ with the best F1.
3) *Evaluation.* The selected model is retrained on the training set and scored on the test fold.

We report the mean and standard deviation of F1 over the test folds. A per-fold difference Δ between two configurations is considered real only if it has the same sign in all five folds. All statistics are learned on training data, and the submitted model is selected by the inner loop on the full training set.

B. Linear models

[MSE gradient descent, SGD (batch size 1), least squares, ridge: losses, grids, iterations, seed; continuous scores thresholded like probabilities.]

C. Logistic models

With labels $y \in \{0, 1\}$, logistic regression minimizes the mean negative log-likelihood $L(w) = \frac{1}{N}\sum_i \big[\log(1 + e^{x_i^\top w}) - y_i x_i^\top w\big]$, computed with `logaddexp` for numerical stability; regularized logistic regression adds $\lambda\|w\|^2$. Both use full-batch gradient descent from $w = 0$ for 300 iterations, after which the loss changes by less than 0.06%. We search $\gamma \in \{0.01, 0.03, 0.1, 0.3, 1\}$, around the convergence bound $\gamma < 0.57$ given by the largest eigenvalue of $X^\top X/N$, and $\lambda \in \{10^{-6}, \ldots, 10^{-1}\}$.

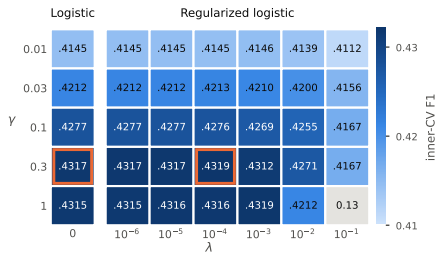


Figure 1. Inner-CV F1 on the full training set; $\lambda = 0$ is plain logistic regression. The best cells differ by less than 0.0004, below the fold standard deviation. Orange: selected. Grey: not converged.

Method	F1	Prec.	Rec.	Acc.	Thr.
GD (MSE)	[]				
SGD (MSE)	[]				
Least squares	[]				
Ridge	[]				
Logistic	0.431 ± 0.005	0.359	0.540	0.874	0.20
Reg. logistic	0.432 ± 0.005	0.358	0.543	0.874	0.20
Final model	**0.441 ± 0.006**	0.369	0.549	0.877	0.21

Table I
NESTED CV RESULTS, MEAN OVER THE 5 OUTER FOLDS. THR.: MEAN SELECTED THRESHOLD.

D. Class imbalance

Only 8.8% of the samples are positive. Besides threshold tuning, we weight the positive samples by $\omega \in \{1, \sqrt{r}, r\}$, where $r = n_-/n_+ \approx 10.3$. For a well-specified model, this multiplies the predicted odds by ω, so a threshold t becomes t' with $t'/(1-t') = \omega\, t/(1-t)$, and weighting should add little to threshold tuning.

E. Final model

A linear model cannot represent the joint effect of two features, for instance that smoking matters more for older respondents. The final model, regularized logistic regression with $\gamma = 0.3$ and $\lambda = 10^{-4}$, is therefore fitted twice: a first fit ranks the features by $|w_j|$, then the 780 products of the 40 top features are standardized and added, and the model is refit. The number 40 was chosen among $\{20, 30, \ldots, 80\}$ by 3-fold cross-validation.

IV. RESULTS

Logistic models. Plain and regularized logistic regression are tied, with $\Delta = 0.0003 \pm 0.0014$. Regularization does not help because the model does not overfit: the training and validation losses are 0.219 and 0.221. Accuracy is not informative, since always predicting "no disease" gives 0.912. **[Person 3: linear vs. logistic comparison once Table I is complete.]**

Features. The codebook rules give the largest gain (Table II), entirely through precision, which rises from 0.338 to 0.359. The indicators and one-hot features add 0.006 in every fold. The interactions add 0.003, but in four folds out of five, so this gain is likely but not certain. On the AIcrowd test set, the final model reaches F1 = 0.450 and accuracy 0.878. Squared, binned or log-transformed features, more interactions, a larger penalty and model averaging changed F1 by less than 0.001.

Configuration	F1	Δ
Generic baseline, step 4 only	0.416 ± 0.005	
+ codebook rules, steps 1–3	0.432 ± 0.005	+0.016 ± 0.002
+ indicators, one-hot	0.438 ± 0.004	+0.006 ± 0.002
+ 40-feature interactions	0.441 ± 0.006	+0.003 ± 0.002
Class weight $\omega = \sqrt{r}$	0.431 ± 0.005	
Class weight $\omega = r$	0.430 ± 0.005	

Table II
ABLATIONS WITH REGULARIZED LOGISTIC REGRESSION ON THE SAME FOLDS. Δ: PER-FOLD GAIN OVER THE ROW ABOVE. THE WEIGHTS APPLY TO THE CODEBOOK MODEL, WHICH IS $\omega = 1$.

Class weighting. It changes F1 by at most 0.0013 but moves the threshold from 0.21 to 0.48 and 0.75, close to the values 0.46 and 0.73 predicted in Section III-D: threshold tuning already handles the imbalance.

V. CONCLUSION

[Final model, its nested-CV F1 and AIcrowd score; main lessons; limitations.]

REPRODUCIBILITY

Code: **[public GitHub URL and commit]**. `run.py` reproduces the submitted predictions; `tune_models.py` defines the tuning grids and runs the nested CV (Table I and the preprocessing rows of Table II), `imbalance.py` the weighting rows, and `plots.py` draws Fig. 1 from the outputs in `results/`.

REFERENCES

[1] Centers for Disease Control and Prevention, "Behavioral risk factor surveillance system: LLCP 2015 codebook report," https://www.cdc.gov/brfss/annual_data/2015/pdf/codebook15_llcp.pdf, 2016, accessed October 2026.